Daak

swarm

Daak and Swarm: the tracks we enter, and what is built

The lid photo of every handoff and every driver's ID photo live on Swarm, encrypted, opened only by a reference nobody public holds

In 60 seconds

1. Every handoff has a photo of the parcel's lid. It goes to Swarm encrypted, so the reference is both the address and the key.
2. That reference never sits in a public record as it is: Daak seals it before it reaches Arkiv. The public log keeps only the photo's hash, so anyone can check a photo shown later is the one taken.
3. When 0G was down, the recheck pulled each photo back from Swarm and asked again. That is how KNP-1204's held pay was released.

Tracks

Arkiv's Swarm bonus
`$500`

Daak enters three partners, the event's limit: Hedera, 0G and Arkiv (with Swarm).

What the track asks, and where Daak stands

Upload encrypted so only the people with the reference can open it	POST /bytes with Swarm-Encrypt: true; a 128-hex reference	Live
/arkiv/swarm.md	What is stored where, an example entity, postage, upload and read code, reproduction steps	Live
The Swarm moment in the video, with its timestamp	To film in the demo run and note in swarm.md	In progress

Every piece, where it lives, and its proof

Lid photos	Uploaded after the driver has the verdict; the hash also goes on HCS `web/lib/swarm.ts, web/app/api/handoff`
Sealed references	AES-256-GCM, a key derived from the server secret and the purpose (sealRef, openRef) `web/lib/swarm.ts`
Operator view	"View lid photo" in /ops unseals the reference on the server and streams the photo from Swarm `web/app/api/photo/[code]/[hop]`

0G recheck	The recheck downloads the encrypted photo and sends it to the sealed model `web/app/api/recheck`	
ID photos	Uploaded encrypted when a driver joins; the reference sits inside the sealed kyc column `web/app/api/join, web/app/api/kyc/[uid]`	
Example	KNP-6025 hop 1: Arkiv entity 0x1d158aeb...31f1, its photo column a sealed reference of 208 base64 characters	arkiv/swarm.md

Honest limits

- Uploads go through the public gateway, which stamps them with its own postage batch; Daak has not bought a batch (that needs xBZZ on Gnosis), so we cannot quote a batch id or promise how long photos stay.
- Sharing is all or nothing: whoever holds a reference can open the photo, so Daak keeps the keys on its server.

To ask the mentor

- For dispute evidence kept 30 to 90 days, what batch depth and duration would you buy, and is pinning needed on top?
- Would ACT (access control) fit for sharing a photo with the shop and the operator but nobody else, instead of server sealing?
- Is there a way to read the batch id behind a gateway upload, for the record?